LUMPING AND THE MIXING TIME OF REVERSIBLE CHAINS WITH n STATES

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Abstract. Aldous and Fill asked whether lumping states of a reversible Markov chain can increase the stopping-rule mixing time $\tau_1^{(2)}$ by more than an absolute constant factor. Hermon and Peres showed that for random walks on graphs of bounded degree the factor can be of the order of the logarithm of the number of vertices, and no larger. For reversible chains with $n \geq 4$ states we show that the largest possible factor $K(n)$ satisfies $\lfloor n/2 \rfloor \leq K(n) \leq n-2$, and that $K(n) = \lfloor n/2 \rfloor$ for $n \leq 6$ and for $n = 8$. The upper bound rests on two facts: lumping two states multiplies $\tau_1^{(2)}$ by less than 2, a constant that cannot be lowered, and the relaxation time is at most $\tau_1^{(2)}$. The lower bound comes from a random walk on a weighted comb. We also show that lumping increases the mean hitting time of a set A by a factor of at most $n - |A|$, and that this bound is sharp up to a factor that tends to 2 as $n \to \infty$.

1. Introduction

Let I be a finite set with $n \geq 2$ elements and let $(w_{xy})_{x,y \in I}$ be a symmetric array of nonnegative weights, loops $w_{xx} > 0$ being allowed, such that the graph with edges $\{x, y\}$, $x \neq y$, $w_{xy} > 0$, is connected. Put $w_x = \sum_y w_{xy}$ and $w = \sum_x w_x$. The random walk on this network is the Markov chain X on I with transition probabilities $p_{xy} = w_{xy}/w_x$; it is reversible with stationary distribution $\pi_x = w_x/w$, and every irreducible reversible chain on I is of this form. We write $T_y = \inf\{t \geq 0 : X_t = y\}$ for the hitting time of a state y and $T_A = \inf\{t \geq 0 : X_t \in A\}$ for the hitting time of a set A, so that $T_y = 0$ for the walk started at y, and $\mathbb{E}_\mu$ and $\mathbb{P}_\mu$ for expectation and probability for the chain started from the distribution μ.

Given a map f from I onto a set $\widehat{I}$, the *induced chain* $\widehat{X}$ is the random walk on $\widehat{I}$ with weights

$$\widehat{w}_{\hat{x}\hat{y}} = \sum_{x \in f^{-1}(\hat{x})} \sum_{y \in f^{-1}(\hat{y})} w_{xy}, \qquad \hat{x}, \hat{y} \in \widehat{I}.$$

Its transition probabilities are $\mathbb{P}_\pi(f(X_1) = \hat{y} \mid f(X_0) = \hat{x})$ and its stationary distribution is $\widehat{\pi}_{\hat{x}} = \pi(f^{-1}(\hat{x}))$; this is the chain of Aldous and Fill [2, Section 4.6.1]. In the language of networks, the states with the same value of f are glued together. We say that $\widehat{X}$ is obtained from X by lumping, and write $\widehat{\mathbb{E}}$ and $\widehat{T}$ for its expectations and hitting times. If g maps $\widehat{I}$ onto a third set, the chain induced from $\widehat{X}$ by g is the chain induced from X by $g \circ f$, as both have the weights $\sum w_{xy}$ over the preimages; so every lumping onto q states of a chain with m states is a composition of $m - q$ lumpings of two states.

The parameter of [2, Chapter 4] that we study is

$$\tau_1^{(2)} = \max_{i \in I} \min_{U} \mathbb{E}_i U,$$

Date: November 2026.

2020 *Mathematics Subject Classification.* 60J10, 05C81.

Key words and phrases. reversible Markov chain, lumping, mixing time, relaxation time, hitting time.

n	2	3	4	5	6	7	8	9	10	11	12
$\lfloor n/2 \rfloor$	1	1	2	2	3	3	4	4	5	5	6
$U(n)$	1	1	2	2	3	4	4	5	6	7	8

TABLE 1. The bounds of Theorem 1 for small n.

where the minimum is over all stopping times U, possibly randomized with a randomization independent of the future of the chain, such that X_U has distribution π under $\mathbb{P}_i$. Lovász and Winkler [4] showed that the minimum is attained and found its value (see also [2, Theorem 9.4]), so that

$$\tau_1^{(2)} = \max_{i,j \in I} \bigl(\mathbb{E}_i T_j - \mathbb{E}_\pi T_j \bigr). \tag{1}$$

For reversible chains $\tau_1^{(2)}$ is equivalent, up to absolute constants, to the total variation mixing time of the chain in continuous time [1], [2, Theorem 4.6]; see also [5].

The maximal mean commute time $\tau^* = \max_{i,j}(\mathbb{E}_i T_j + \mathbb{E}_j T_i)$, the average hitting time $\tau_0 = \sum_j \pi_j \mathbb{E}_i T_j$ (which does not depend on i), the relaxation time $\tau_2 = 1/(1-\lambda_2)$, where λ_2 is the second largest eigenvalue of (p_{xy}), and the Cheeger time constant τ_c do not increase under lumping [2, Proposition 4.44]. For the mixing time the matter was left open:

Open Problem 4.45 of [2]**.** *Is the value of $\tau_1^{(2)}$ in an induced chain bounded by K times the value of $\tau_1^{(2)}$ in the original chain, for some absolute constant K? For $K = 1$?*

Hermon and Peres [3, Theorem 3(c)] answered both questions in the negative. They construct graphs G_n of bounded degree for which lumping together certain pairs of vertices increases the total variation mixing time of the lazy simple random walk by a factor of order $\log |V(G_n)|$. Lumping does not increase the relaxation time, and the mixing time of a lazy reversible chain is bounded below by a constant multiple of the relaxation time and above by a constant multiple of the relaxation time multiplied by $\log(1/\min_x \pi_x)$; hence this is the largest possible order in that setting [3, Remark 1.8].

For a random walk on a weighted network $\min_x \pi_x$ can be arbitrarily small, and the bound in terms of $\log(1/\min_x \pi_x)$ says nothing about the dependence on the number of states. This paper determines that dependence up to a factor that tends to two as $n \to \infty$. Let

$$K(n) = \sup \ \frac{\tau_1^{(2)}(\widehat{X})}{\tau_1^{(2)}(X)},$$

the supremum being over all reversible chains X with at least two and at most n states and all induced chains $\widehat{X}$ of X. Put $U(2) = 1$ and

$$U(n) = \max_{2 \le q \le n-1} \ \min\bigl(2^{\,n-q},\ q-1\bigr) \qquad (n \ge 3).$$

Theorem 1. *For every $n \ge 2$,*

$$\left\lfloor \frac{n}{2} \right\rfloor \ \le \ K(n) \ \le \ U(n) \ \le \ \max(1, n-2).$$

In particular $K(n) = \lfloor n/2 \rfloor$ for $2 \le n \le 6$ and for $n = 8$, and $3 \le K(7) \le 4$.

The upper bound combines two inequalities. The first concerns a single lumping of two states.

Theorem 2. *Let $\widehat{X}$ be obtained from X by lumping two states and keeping the other states apart. Then $\tau_1^{(2)}(\widehat{X}) < 2\,\tau_1^{(2)}(X)$, and the constant 2 cannot be replaced by a smaller one.*

Applied $m-q$ times, Theorem 2 bounds the factor by 2^{m-q} for a lumping of a chain with m states onto q states. The second inequality bounds the factor by $q-1$; it is a chain of four inequalities, three of which are in [2]. The fourth is the following comparison of the relaxation time with the stopping-rule mixing time, in which the constant is one; whether this holds is asked in [2, Open Problem 4.48(b)]. Comparisons of the two quantities up to absolute constants are in [1, 5].

Lemma 3. *Let $\lambda < 1$ be a real eigenvalue of the transition matrix of an irreducible chain on a finite set, with a real eigenfunction g, and let i be a state at which $|g|$ is largest. If U is a stopping time such that X_U has the stationary distribution π under $\mathbb{P}_i$, then $\mathbb{E}_i U \geq 1/(1-\lambda)$. In particular $\tau_2 \leq \tau_1^{(2)}$ for every reversible chain.*

Equality $\tau_2 = \tau_1^{(2)}$ holds for every chain with two states and for the simple random walk on a complete graph.

The lower bound in Theorem 1 is approached along a family of random walks on a weighted comb with $2N+4$ states, described in Section 5. Lumping each of N teeth with a vertex of the spine, and the end of the spine with a heavy leaf, multiplies $\tau_1^{(2)}$ by a factor that tends to $N+2$. With $N=0$ only two states are lumped and the factor tends to 2, which shows that the constant in Theorem 2 is optimal. In these examples the stationary distribution is very far from uniform, in agreement with the bound of [3]; see Remark 9.

The comb uses $N+1$ lumpings of two states and reaches the factor $N+2$. We do not know whether a composition of several lumpings of two states can multiply $\tau_1^{(2)}$ by more than one plus their number. If they cannot, the argument of Section 4 bounds the factor for a lumping onto q states by $\min(m-q+1, q-1) \leq \lfloor m/2 \rfloor$, and then $K(n) = \lfloor n/2 \rfloor$ for every n.

The mechanism behind the lower bound is a one-sided hitting time. Aldous and Fill observe [2, Section 4.6.1] that, although commute times decrease under lumping, a mean hitting time may increase, and give an example on four vertices where it doubles. The next proposition bounds the increase for hitting times of sets.

Proposition 4. *Let $\widehat{X}$ be the chain induced from X by $f : I \to \widehat{I}$, let $\widehat{A}$ be a non-empty proper subset of $\widehat{I}$ and $A = f^{-1}(\widehat{A})$. Then for every $\hat{x} \in \widehat{I} \setminus \widehat{A}$*

$$\widehat{\mathbb{E}}_{\hat{x}} \widehat{T}_{\widehat{A}} \;\leq\; \sum_{y \in I \setminus A} \mathbb{E}_y T_A \;\leq\; (n - |A|) \max_{y \in I} \mathbb{E}_y T_A.$$

For every $N \geq 0$ and $\varepsilon > 0$ there is a chain with $n = 2N+4$ states, an induced chain and a set A with two elements such that $\widehat{\mathbb{E}}_{\hat{x}} \widehat{T}_{\widehat{A}} \geq (N+1-\varepsilon) \max_y \mathbb{E}_y T_A$ for some $\hat{x}$.

Section 2 collects identities for hitting times, Theorem 2 is proved in Section 3, Lemma 3 and the upper bound of Theorem 1 in Section 4, the lower bound in Section 5 and Proposition 4 in Section 6.

2. Hitting-time potentials

Throughout, $H(x,j) = \mathbb{E}_x T_j$ and $F_{ij} = \mathbb{E}_i T_j - \mathbb{E}_\pi T_j$, so that $\tau_1^{(2)} = \max_{i,j} F_{ij}$ by (1). Functions on I are column vectors; ω is the vector $(w_x)_{x \in I}$, $\mathbf{1}$ the constant vector 1 and δ_x the indicator of x. The Laplacian L has entries $L_{xy} = -w_{xy}$ for $x \neq y$ and $L_{xx} = \sum_{y \neq x} w_{xy}$, so that loops do not enter it, and the Dirichlet form is

$$\mathcal{E}(g, g') = g^{\mathsf{T}} L g' = \frac{1}{2} \sum_{x,y} w_{xy} \, (g_x - g_y)(g'_x - g'_y).$$

The kernel of L consists of the constants, and $\mathbf{1}^{\mathsf{T}}Lg = 0$ for every g. For a state y and a set Z not containing it, $R(y \leftrightarrow Z)$ is the effective resistance between y and Z in the network with conductances w_{xy}, $x \neq y$. For two states $u \neq v$ the function $h_x = \mathbb{P}_x(T_u < T_v)$ satisfies [6, Section 2.2]

$$Lh = \frac{\delta_u - \delta_v}{R(u \leftrightarrow v)}, \qquad \mathcal{E}(h,h) = \frac{1}{R(u \leftrightarrow v)}. \tag{2}$$

Fix a state o. Since the network is connected, the matrix obtained from L by deleting the row and the column of o is invertible; let Γ be its inverse, extended by zeros in the row and the column of o. Then Γ is symmetric and $L\Gamma\delta_j = \delta_j - \delta_o$ for $j \neq o$, so that $\Gamma\delta_j$ is the voltage, vanishing at o, of the unit current flow from j to o. Put $\zeta = \Gamma\omega$ and $\bar{\zeta} = \sum_x \pi_x \zeta_x$.

Lemma 5. (i) *$H(i,j) = \zeta_i - \zeta_j - w(\Gamma_{ij} - \Gamma_{jj})$ for all i, j; in particular $\zeta_i = H(i,o)$.*

(ii) *$F_{ij} = \zeta_i + \zeta_j - \bar{\zeta} - w\Gamma_{ij}$. In particular F is symmetric, $\sum_x \pi_x F_{ix} = 0$, and $LF\delta_j = \omega - w\delta_j$.*

(iii) *For $i \in I$ let $\kappa_i = \max_j F_{ij}$ and let φ_i be the function $\varphi_i(x) = \kappa_i - F_{ix}$. Then $\kappa_i \geq 0$, $\varphi_i \geq 0$, $\sum_x \pi_x \varphi_i(x) = \kappa_i$, $\tau_1^{(2)} = \max_i \kappa_i$, and for every function g on I*

$$\mathcal{E}(\varphi_i, g) = w\Big(g_i - \sum_x \pi_x g_x\Big), \qquad \mathcal{E}(\varphi_i, \varphi_j) = -wF_{ij}. \tag{3}$$

Proof. (i) For $x \neq j$, conditioning on the first step gives $w_x H(x,j) = w_x + \sum_y w_{xy} H(y,j)$, that is, $(LH(\cdot,j))_x = w_x$; as the entries of $LH(\cdot,j)$ sum to zero, $(LH(\cdot,j))_j = w_j - w$. Thus $LH(\cdot,j) = \omega - w\delta_j$ and $H(j,j) = 0$, and these two conditions determine $H(\cdot,j)$. Now $L\zeta = \sum_{j \neq o} w_j(\delta_j - \delta_o) = \omega - w\delta_o$, so the right side of (i), as a function of i, has Laplacian $\omega - w\delta_o - w(\delta_j - \delta_o) = \omega - w\delta_j$ when $j \neq o$ and $\omega - w\delta_o$ when $j = o$, and it vanishes at $i = j$. This proves the formula, and $\zeta_o = 0$, $\Gamma_{io} = 0$ give $\zeta_i = H(i,o)$.

(ii) By the symmetry of Γ, $\sum_i \pi_i \Gamma_{ij} = \zeta_j / w$. Averaging (i) over i with weights π_i gives $\mathbb{E}_\pi T_j = \bar{\zeta} - 2\zeta_j + w\Gamma_{jj}$, and subtracting this from (i) gives the formula for F_{ij}, which is symmetric. Each column of F has π-mean zero by definition, hence so does each row, and $LF\delta_j = LH(\cdot,j)$.

(iii) A row of F has π-mean zero, so its maximum κ_i is nonnegative, and $\varphi_i \geq 0$ has π-mean κ_i. The identity $\tau_1^{(2)} = \max_i \kappa_i$ is (1). By symmetry of F and (ii), $L\varphi_i = -LF\delta_i = w\delta_i - \omega$, so $\mathcal{E}(\varphi_i, g) = g^{\mathsf{T}} L\varphi_i = wg_i - \sum_x w_x g_x$. With $g = \varphi_j$ this is $w(\kappa_j - F_{ji} - \kappa_j) = -wF_{ij}$. □

We also use two facts about networks whose graph, loops removed, is a tree.

Lemma 6. *Let $x \neq y$ be states with $w_{xy} > 0$ such that every path from x to y in the network uses the edge $\{x,y\}$, and let V_{xy} be the set of states that can be reached from x without visiting y. Then $\mathbb{E}_x T_y = w(V_{xy})/w_{xy}$, where $w(V_{xy}) = \sum_{z \in V_{xy}} w_z$.*

Proof. Up to time T_y the walk started at x stays in V_{xy}, so $\mathbb{E}_x T_y$ is the same for the network on $V_{xy} \cup \{y\}$ in which y keeps only the edge to x. There the stationary probability of y is $w_{xy}/(w(V_{xy}) + w_{xy})$, so the mean return time to y is $1 + w(V_{xy})/w_{xy}$, and it also equals $1 + \mathbb{E}_x T_y$. □

Lemma 7. *Suppose that the graph of the network, loops removed, is a tree. For a state x let $[x,o]$ be the set of edges of the path from x to o, for an edge e let w_e be its weight, and let S_e be the total weight $\sum_z w_z$ of the states separated from o by e. Then*

$$\Gamma_{ij} = \sum_{e \in [i,o] \cap [j,o]} \frac{1}{w_e}, \qquad H(x,o) = \sum_{e \in [x,o]} \frac{S_e}{w_e}, \qquad \mathbb{E}_\pi T_o = \frac{1}{w} \sum_e \frac{S_e^2}{w_e},$$

the last sum being over all edges of the tree.

Proof. On a tree the unit current flow from j to o runs along the path from j to o, and its voltage at i is the sum of the resistances $1/w_e$ of the edges of that path through which the path from i to o also passes. The walk from x to o visits every state of the path between them in turn, and by Lemma 6 the edge e of that path is crossed towards o in mean time S_e/w_e. Averaging over x with weights π_x, the edge e is counted with total weight S_e/w. □

3. Lumping two states

Let $u \neq v$ be states, let f send u and v to one state $\hat{v}$, send no other state to $\hat{v}$ and be injective on $J = I \setminus \{u, v\}$, and let $\widehat{F}$ be the matrix F of $\widehat{X}$. Let $h_x = \mathbb{P}_x(T_u < T_v)$, $R = R(u \leftrightarrow v)$ and $\bar{h} = \sum_x \pi_x h_x$, and for $i \in I$ put

$$\beta_i = \varphi_i(u) - \varphi_i(v) = F_{iv} - F_{iu}.$$

Proposition 8. *For all $i, j \in I$,*

$$\beta_i = wR\,(h_i - \bar{h}), \qquad \widehat{F}_{f(i)f(j)} = F_{ij} + \frac{\beta_i \beta_j}{wR},$$

and

$$\widehat{F}_{f(i)f(j)} \;\leq\; \kappa_i + \kappa_j. \tag{4}$$

Proof. Take $o = v$ in Section 2. By (2) the voltage of the unit current flow from u to v is Rh, so $\Gamma\delta_u = Rh$; in particular $\Gamma_{uu} = R$, $\zeta_u = \sum_x \Gamma_{ux} w_x = wR\bar{h}$ and $\zeta_v = 0$. Lemma 5(ii) gives $\beta_i = \zeta_v - \zeta_u + w(\Gamma_{iu} - \Gamma_{iv}) = wR(h_i - \bar{h})$.

Let $\widetilde{\Gamma} = \Gamma - Rhh^{\mathsf{T}}$. Its rows and columns at u and v vanish. The rows of $L\Gamma$ indexed by J, restricted to the columns indexed by J and by u, give $L_{JJ}\Gamma_{JJ} + L_{Ju}\Gamma_{uJ} = I_J$ and $L_{JJ}\Gamma_{Ju} + L_{Ju}\Gamma_{uu} = 0$, where the subscripts denote submatrices; hence $L_{JJ}(\Gamma_{JJ} - \Gamma_{Ju}\Gamma_{uJ}/\Gamma_{uu}) = I_J$, that is, $\widetilde{\Gamma}_{JJ} = L_{JJ}^{-1}$. The Laplacian of $\widehat{X}$ with the row and the column of $\hat{v}$ deleted is L_{JJ}, the total weight of $\widehat{X}$ is w, and $\widehat{\pi}_{f(x)} = \pi_x$ for $x \in J$, $\widehat{\pi}_{\hat{v}} = \pi_u + \pi_v$. So the matrix Γ of $\widehat{X}$ grounded at $\hat{v}$ has entries $\widetilde{\Gamma}_{ij}$ at $(f(i), f(j))$.

For a matrix A indexed by I write $\mathcal{C}(A)_{ij} = A_{ij} - (A\pi)_i - (\pi^{\mathsf{T}}A)_j + \pi^{\mathsf{T}}A\pi$. Lemma 5(ii) says that $F = -w\,\mathcal{C}(\Gamma)$. If A_{ij} depends only on $f(i)$ and $f(j)$, then $\mathcal{C}(A)_{ij}$ equals the same expression formed with the quotient matrix and $\widehat{\pi}$ at $(f(i), f(j))$. Applied to $\widetilde{\Gamma}$, and since $\mathcal{C}(hh^{\mathsf{T}})_{ij} = (h_i - \bar{h})(h_j - \bar{h})$,

$$\widehat{F}_{f(i)f(j)} = -w\,\mathcal{C}(\widetilde{\Gamma})_{ij} = F_{ij} + wR\,(h_i - \bar{h})(h_j - \bar{h}) = F_{ij} + \frac{\beta_i\beta_j}{wR}.$$

If $\beta_i\beta_j \leq 0$ this gives $\widehat{F}_{f(i)f(j)} \leq F_{ij} \leq \kappa_i \leq \kappa_i + \kappa_j$. Suppose that $\beta_i\beta_j > 0$. Exchanging the names of u and v changes the signs of β_i and β_j together and leaves the statement unchanged, so we may assume $\beta_i, \beta_j > 0$. Put $g_i = \varphi_i/\beta_i$, $g_j = \varphi_j/\beta_j$ and let $\eta = \min(g_i, g_j)$ pointwise. These functions are nonnegative, and $g_i(u) - g_i(v) = g_j(u) - g_j(v) = 1$, hence $\eta(u) - \eta(v) = 1$. The function $\eta - \eta(v)\mathbf{1} - h$ vanishes at u and at v, so by (2) it is $\mathcal{E}$-orthogonal to h, and

$$\mathcal{E}(\eta, \eta) \;\geq\; \mathcal{E}(h, h) = \frac{1}{R}.$$

Let $\xi_i = g_i - \eta$ and $\xi_j = g_j - \eta$. They are nonnegative and at every state at least one of them vanishes, so $\mathcal{E}(\xi_i, \xi_j) = -\sum_{x \neq y} w_{xy}\xi_i(x)\xi_j(y) \leq 0$. Expanding $g_i = \eta + \xi_i$ and $g_j = \eta + \xi_j$,

$$\mathcal{E}(g_i, \eta) + \mathcal{E}(g_j, \eta) - \mathcal{E}(g_i, g_j) = \mathcal{E}(\eta, \eta) - \mathcal{E}(\xi_i, \xi_j) \;\geq\; \frac{1}{R}.$$

Let $\bar\eta = \sum_x \pi_x \eta(x) \ge 0$. By (3) the left side equals

$$\frac{w}{\beta_i}\big(\eta(i)-\bar\eta\big) + \frac{w}{\beta_j}\big(\eta(j)-\bar\eta\big) + \frac{wF_{ij}}{\beta_i\beta_j},$$

and $\eta(i) \le g_j(i) = (\kappa_j - F_{ij})/\beta_j$, $\eta(j) \le g_i(j) = (\kappa_i - F_{ij})/\beta_i$. Hence

$$\frac{1}{R} \;\le\; \frac{w}{\beta_i\beta_j}\big(\kappa_i + \kappa_j - F_{ij} - (\beta_i+\beta_j)\bar\eta\big),$$

and multiplying by $\beta_i\beta_j/w$ and using the formula for $\widehat F$,

$$\widehat F_{f(i)f(j)} \;\le\; \kappa_i + \kappa_j - (\beta_i+\beta_j)\,\bar\eta. \tag{5}$$ □

Proof of Theorem 2. By (1) and Lemma 5(iii), $\tau_1^{(2)}(\widehat X)$ is the largest entry of $\widehat F$, and every entry is $\widehat F_{f(i)f(j)}$ for some $i,j \in I$. If $\beta_i\beta_j \le 0$ that entry is at most $\kappa_i \le \tau_1^{(2)}(X)$. Otherwise, in the notation of the proof above, $\eta(u) = \eta(v) + 1 \ge 1$, so $\bar\eta \ge \pi_u > 0$, and (5) gives $\widehat F_{f(i)f(j)} < \kappa_i + \kappa_j \le 2\tau_1^{(2)}(X)$. Finally $\tau_1^{(2)}(X) > 0$: otherwise every row of F, having π-mean zero, would vanish, against $LF\delta_j = \omega - w\delta_j \ne 0$. As there are finitely many entries, $\tau_1^{(2)}(\widehat X) < 2\tau_1^{(2)}(X)$. That the constant 2 is optimal is shown by the comb of Section 5 with $N = 0$ (Proposition 10). □

4. The upper bound

Proof of Lemma 3. Replacing g by a multiple we may assume that $g(i) = 1 = \max_x |g(x)|$. Since $Pg = \lambda g$, where $P = (p_{xy})$, the process

$$\mathcal{M}_t = g(X_t) + (1-\lambda)\sum_{s<t} g(X_s), \qquad t \ge 0,$$

is a martingale with $\mathcal{M}_0 = 1$ under $\mathbb{P}_i$, with respect to the filtration generated by the chain and by the randomization used by U. We may assume that $\mathbb{E}_i U < \infty$. The increments of $\mathcal{M}$ are bounded by 2, so the optional stopping theorem gives $\mathbb{E}_i \mathcal{M}_U = 1$. As $\lambda \ne 1$, $\sum_x \pi_x g(x) = \sum_x \pi_x (Pg)(x) = \lambda \sum_x \pi_x g(x)$ forces $\sum_x \pi_x g(x) = 0$. Hence $\mathbb{E}_i g(X_U) = 0$ and

$$1 = (1-\lambda)\,\mathbb{E}_i \sum_{s<U} g(X_s) \le (1-\lambda)\,\mathbb{E}_i U.$$

For a reversible chain all eigenvalues are real; taking $\lambda = \lambda_2$ gives $\tau_1^{(2)} \ge \min_U \mathbb{E}_i U \ge \tau_2$. □

Proof of the upper bound in Theorem 1. Let X have $m \le n$ states and let $\widehat X$ be an induced chain with q states. If $q = m$ the map f is a bijection and $\widehat X$ is X with its states renamed, and if $q = 1$ then $\tau_1^{(2)}(\widehat X) = 0$. Otherwise $m \ge 3$ and $2 \le q \le m-1$. The lumping is a composition of $m - q$ lumpings of two states, so Theorem 2 gives $\tau_1^{(2)}(\widehat X) \le 2^{m-q}\tau_1^{(2)}(X)$. We also have

$$\tau_1^{(2)}(\widehat X) \;\le\; \tau_0(\widehat X) \;\le\; (q-1)\,\tau_2(\widehat X) \;\le\; (q-1)\,\tau_2(X) \;\le\; (q-1)\,\tau_1^{(2)}(X). \tag{6}$$

For the first inequality, choose a state J of $\widehat X$ with distribution $\widehat\pi$, independently of the chain, and stop at the hitting time of J; the chain is then in the stationary distribution, and the mean of this stopping time is $\tau_0(\widehat X)$ from every starting state [2, Section 4.3]. The second is the eigentime identity $\tau_0 = \sum_{k\ge 2}(1-\lambda_k)^{-1}$ [2, Proposition 3.13 and Lemma 4.24] for the reversible chain $\widehat X$, each of whose $q-1$ terms is at most $\tau_2(\widehat X)$. The third is part of the contraction principle [2, Proposition 4.44]: a function g on $\widehat I$ and the function $g \circ f$ on I have the same mean, variance and Dirichlet form under $\widehat\pi$ and π, so the spectral gap of $\widehat X$ is at

least that of X. The fourth is Lemma 3. Hence the ratio is at most $\min(2^{m-q}, q-1) \le U(m)$. Finally $U(m) \le U(n)$, because $2^{m-q} \le 2^{n-q}$, and $U(n) \le \max(1, n-2)$, because each term is at most $q-1 \le n-2$. □

Remark 9. Write $\pi_* = \min_x \pi_x$. In discrete time $\tau_1^{(2)} \le \frac{4e}{e-1}\,\tau_2\,(1 + \frac{1}{2}\log(1/\pi_*))$ by [2, Lemma 4.23]. Since $\tau_2(\widehat{X}) \le \tau_2(X)$ and $\min_{\hat{x}} \widehat{\pi}_{\hat{x}} \ge \pi_*$, Lemma 3 gives

$$\tau_1^{(2)}(\widehat{X}) \;\le\; \frac{4e}{e-1}\Big(1 + \frac{1}{2}\log\frac{1}{\pi_*}\Big)\tau_1^{(2)}(X),$$

which is the form taken for $\tau_1^{(2)}$ by the bound of Hermon and Peres [3, Remark 1.8]. Together with Theorem 1, the factor is at most $\min\big(U(n),\ \frac{4e}{e-1}(1+\frac{1}{2}\log(1/\pi_*))\big)$. For the simple random walk on a graph with bounded degrees $\log(1/\pi_*)$ is of order $\log n$, and the examples of [3] show that this order is attained. In the comb of Section 5, $\log(1/\pi_*)$ is of order $(N+1)\log t$.

5. The comb

Fix an integer $N \ge 0$ and a real number $t \ge 2$, and put $r = t^3$ and $\alpha = t$. The *comb* is the network on the $2N+4$ states a, b, c, $z_0, \dots, z_N$, $y_0, \dots, y_{N-1}$ with the edge weights

$$w_{az_0} = 1, \qquad w_{bz_N} = \alpha r^{N+1}, \qquad w_{cz_N} = r^{N+2},$$
$$w_{z_k z_{k+1}} = r^{k+1}, \qquad w_{y_k z_{k+1}} = \alpha r^{k+1} \qquad (0 \le k < N),$$

the loop weights

$$w_{aa} = t-1, \qquad w_{bb} = (t-1)\alpha r^{N+1}, \qquad w_{cc} = \ell = (N+2)\,r^{N+4},$$
$$w_{y_k y_k} = (t-1)\alpha r^{k+1} \qquad (0 \le k < N),$$

and all other weights zero. Without its loops the comb is a tree: the spine $a, z_0, \dots, z_N, c$, the teeth y_k hanging from z_{k+1}, and the leaf b hanging from z_N. From a, from b and from each tooth the walk leaves with probability $1/t$ at each step. The map f sends z_k and y_k to one state $\hat{z}_k$ for $0 \le k < N$, sends z_N and c to one state $\hat{c}$, and keeps a and b. So $\widehat{X}$ has $N+3$ states and is obtained by $N+1$ lumpings of two states. For $N = 0$ the comb is a star with centre z_0, and only z_0 and c are lumped.

Put $\sigma_k = \sum_{l=1}^{k} r^l$ and $B = 2 + \alpha(t+1)$, and

$$S_k = t + 1 + r^{k+1} + B\sigma_k, \qquad \widehat{S}_k = S_k + t\alpha r^{k+1} \qquad (0 \le k < N),$$
$$S_N = t + 1 + r^{N+2} + B\sigma_N + (t+1)\alpha r^{N+1},$$
$$M = t + \sum_{k=0}^{N-1} \frac{S_k}{r^{k+1}} + \frac{S_N}{r^{N+2}}, \qquad \widehat{M} = t + \sum_{k=0}^{N-1} \frac{\widehat{S}_k}{(1+\alpha)r^{k+1}}.$$

Proposition 10. *For the comb X and its induced chain $\widehat{X}$:*

(i) *S_k is the total weight of the states separated from c by the edge $\{z_k, z_{k+1}\}$ ($k < N$), S_N is that of the states other than c, and $w = \ell + r^{N+2} + S_N$. Moreover $M = \mathbb{E}_a T_c = \max_x \mathbb{E}_x T_c$, $\widehat{M} = \widehat{\mathbb{E}}_a \widehat{T}_{\hat{c}}$, $\widehat{\mathbb{E}}_b \widehat{T}_{\hat{c}} = t$, and $\widehat{\mathbb{E}}_{\hat{x}} \widehat{T}_{\hat{c}} \le \widehat{M}$ for $\hat{x} \ne b$.*

(ii) *If $w \ge M r^{N+2}$, then $\tau_1^{(2)}(X) = M - \mathbb{E}_\pi T_c$. If moreover $N = 0$ or $w \ge (\widehat{M} - t)(1+\alpha)r^N$, then $\tau_1^{(2)}(\widehat{X}) = \widehat{M} + t - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$.*

(iii) *As $t \to \infty$ with N fixed, the conditions of (ii) hold, $M = t + N + 1 + O(1/t)$, $\widehat{M} = (N+1)t + O(1)$, and $\mathbb{E}_\pi T_c$ and $\widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$ tend to zero. Hence*

$$\frac{\tau_1^{(2)}(\widehat{X})}{\tau_1^{(2)}(X)} \to N+2, \qquad \frac{\widehat{M}}{M} \to N+1.$$

Proof. (i) The state weights are $w_a = t$, $w_{y_k} = t\alpha r^{k+1}$, $w_b = t\alpha r^{N+1}$, $w_c = r^{N+2} + \ell$, and for the spine $w_{z_0} = 1 + r$ and $w_{z_k} = r^k + r^{k+1} + \alpha r^k$ $(1 \le k < N)$, $w_{z_N} = r^N + \alpha r^N + \alpha r^{N+1} + r^{N+2}$ when $N \ge 1$, while $w_{z_0} = 1 + \alpha r + r^2$ when $N = 0$. The states separated from c by $\{z_k, z_{k+1}\}$ are a, $z_0, \dots, z_k$ and $y_0, \dots, y_{k-1}$. Their total weight is $t + 1 + r$ for $k = 0$, and passing from $k-1$ to k adds $w_{z_k} + w_{y_{k-1}} = r^{k+1} + (B-1)r^k$, which is also the increase of $t + 1 + r^{k+1} + B\sigma_k$. The same count with w_b added gives S_N, and $w = w_c + S_N$.

By Lemma 6, $\mathbb{E}_a T_{z_0} = t$, $\mathbb{E}_{z_k} T_{z_{k+1}} = S_k / r^{k+1}$, $\mathbb{E}_{z_N} T_c = S_N / r^{N+2}$, $\mathbb{E}_{y_k} T_{z_{k+1}} = t$ and $\mathbb{E}_b T_{z_N} = t$. The walk from x to c visits every state of the path between them, so $\mathbb{E}_x T_c$ is the sum of these terms along the path. From a this sum is M. From any other state it is either a sum of some of the spine terms, or t followed by some of the spine terms, and in both cases at most M.

The induced chain is again a tree: the path $a, \hat{z}_0, \dots, \hat{z}_{N-1}, \hat{c}$ with the leaf b attached to $\hat{c}$. Writing $\hat{z}_N = \hat{c}$, the edge $\{\hat{z}_k, \hat{z}_{k+1}\}$ has weight $w_{z_k z_{k+1}} + w_{y_k z_{k+1}} = (1+\alpha) r^{k+1}$, and the states separated from $\hat{c}$ by it are $a, \hat{z}_0, \dots, \hat{z}_k$, of total weight $S_k + w_{y_k} = \widehat{S}_k$. The edge $\{a, \hat{z}_0\}$ (or $\{a, \hat{c}\}$ if $N = 0$) has weight 1 and $\{b, \hat{c}\}$ has weight αr^{N+1}. Lemma 6 and the argument above give the assertions about $\widehat{X}$.

(ii) Take $o = c$ in Section 2. By Lemma 5, $F_{ij} = H(i,c) + H(j,c) - \mathbb{E}_\pi T_c - w\Gamma_{ij}$. For $j = c$ this is $H(i,c) - \mathbb{E}_\pi T_c \le M - \mathbb{E}_\pi T_c$, with equality at $i = a$. For $i, j \ne c$ the edge $\{z_N, c\}$ lies in $[i,c] \cap [j,c]$, so $\Gamma_{ij} \ge r^{-(N+2)}$ by Lemma 7, and $F_{ij} \le 2M - \mathbb{E}_\pi T_c - w r^{-(N+2)} \le M - \mathbb{E}_\pi T_c$. By (1), $\tau_1^{(2)}(X) = M - \mathbb{E}_\pi T_c$.

In $\widehat{X}$ take $o = \hat{c}$ and write $\widehat{H}, \widehat{\Gamma}$ for the corresponding quantities. The paths from a and from b to $\hat{c}$ share no edge, so $\widehat{F}_{ab} = \widehat{M} + t - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$. Every other entry is at most this: entries with $\hat{c}$ are at most $\widehat{M} - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$; entries $\widehat{F}_{b\hat{x}}$ with $\hat{x} \ne b$ are at most $\widehat{H}(\hat{x}, \hat{c}) + t - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$; $\widehat{F}_{bb} \le 2t - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$ and $t \le \widehat{M}$; if $N = 0$ the only remaining entry is $\widehat{F}_{aa} \le 2t - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$ and $\widehat{M} = t$; and if $N \ge 1$, two states of the path other than $\hat{c}$ share the edge $\{\hat{z}_{N-1}, \hat{c}\}$, so $\widehat{\Gamma} \ge 1/((1+\alpha) r^N)$ for them and their entry is at most $2\widehat{M} - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}} - w/((1+\alpha) r^N) \le \widehat{M} + t - \widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}}$.

(iii) Since $\sigma_k \le 2r^k$, $B = t^2 + t + 2$ and $r = t^3$, each of S_k / r^{k+1} $(k < N)$ and S_N / r^{N+2} is $1 + O(1/t)$, so $M = t + N + 1 + O(1/t)$, and $\widehat{S}_k / ((1+\alpha) r^{k+1}) = (t^2 + 1 + O(1/t))/(t+1) = t - 1 + O(1/t)$, so $\widehat{M} = (N+1)t - N + O(1/t)$. As $S_N = O(r^{N+2})$ and $w \ge \ell = (N+2) r^{N+4}$, the conditions of (ii) hold for large t. Next, $\mathbb{E}_\pi T_c \le (1 - \pi_c) M = (S_N / w) M = O(t^{-5})$, and since the states of $\widehat{X}$ other than $\hat{c}$ carry stationary mass at most S_N / w, also $\widehat{\mathbb{E}}_{\widehat{\pi}} \widehat{T}_{\hat{c}} \le (S_N / w) \widehat{M} = O(t^{-5})$. By (ii) the ratio is $((N+2)t - N + o(1))/(t + N + 1 + o(1)) \to N + 2$. □

For $r \gg \alpha t$ a walk on the spine of the comb rarely enters a tooth, and each tooth or leaf delays a walk that starts there by its holding time, of mean t. After lumping, each state $\hat{z}_k$ inherits the holding loop of the tooth y_k, and since α is large the walk from a to $\hat{c}$ is delayed by about t at each of the N levels. In X the far ends a and b are each about t away from c and the heavy state c separates them; in $\widehat{X}$ the delays add up along the spine, and the leaf b adds one more t.

Proof of the lower bound in Theorem 1. The identity map gives $K(n) \ge 1$. For $n \ge 4$ let $N = \lfloor n/2 \rfloor - 2$. The comb has $2N + 4 \le n$ states, and by Proposition 10 the ratio $\tau_1^{(2)}(\widehat{X}) / \tau_1^{(2)}(X)$

N	states	$\tau_1^{(2)}(X)$	$\tau_1^{(2)}(\widehat{X})$	$\tau_1^{(2)}(\widehat{X})/\tau_1^{(2)}(X)$	$\widehat{M}/M$
0	4	1001.001	2000.000	1.9980	0.9990
1	6	1002.001	2999.002	2.9930	1.9950
2	8	1003.002	3998.004	3.9860	2.9890
3	10	1004.003	4997.006	4.9771	3.9811

TABLE 2. The comb with $t = 1000$ and its induced chain.

tends to $N + 2 = \lfloor n/2 \rfloor$ as $t \to \infty$. The stated values of $K(n)$ and the bounds on $K(7)$ follow from Table 1. □

Table 2 gives the values for $t = 1000$, computed in rational arithmetic from (1) and rounded; the closed forms of Proposition 10 agree with them exactly, and the conditions of Proposition 10(ii) hold already for $t = 2$ when $N \le 3$.

6. Hitting times of sets

For a non-empty proper subset Z of I and $x, y \notin Z$ let $G_Z(x, y)$ be the expected number of visits to y before T_Z for the walk started at x. Then [6, Section 2.2]

(7)
$$\mathbb{E}_x T_Z = \sum_{y \notin Z} G_Z(x, y), \qquad G_Z(x, y) = \mathbb{P}_x(T_y < T_Z)\, G_Z(y, y), \qquad G_Z(y, y) = w_y\, R(y \leftrightarrow Z).$$

The formula for $G_Z(y, y)$ holds in the presence of loops: the walk at y reaches Z before returning to y with probability $1/(w_y R(y \leftrightarrow Z))$, and a loop at y changes w_y only.

Proof of Proposition 4. The network of $\widehat{X}$ is obtained from that of X by gluing together the states in each set $f^{-1}(\hat{y})$; edges inside such a set become loops. By Rayleigh's monotonicity principle [6, Section 2.4] gluing does not increase effective resistances, so $\widehat{R}(\hat{y} \leftrightarrow \widehat{A}) \le R(y \leftrightarrow A)$ for every $\hat{y} \in \widehat{I} \setminus \widehat{A}$ and every $y \in f^{-1}(\hat{y})$, where $\widehat{R}$ is the effective resistance in the network of $\widehat{X}$. With (7) applied to both chains,

$$\begin{aligned} \widehat{\mathbb{E}}_{\hat{x}} \widehat{T}_{\widehat{A}} \;&\le\; \sum_{\hat{y} \notin \widehat{A}} \widehat{w}_{\hat{y}}\, \widehat{R}(\hat{y} \leftrightarrow \widehat{A}) = \sum_{\hat{y} \notin \widehat{A}} \sum_{y \in f^{-1}(\hat{y})} w_y\, \widehat{R}(\hat{y} \leftrightarrow \widehat{A}) \\ &\le\; \sum_{y \notin A} w_y\, R(y \leftrightarrow A) = \sum_{y \notin A} G_A(y, y) \;\le\; \sum_{y \notin A} \mathbb{E}_y T_A. \end{aligned}$$

For the second statement take the comb of Section 5, $\widehat{A} = \{\hat{c}\}$, $A = \{z_N, c\}$ and $\hat{x} = a$. Then $\max_y \mathbb{E}_y T_A \le \max_y \mathbb{E}_y T_c = M$ and $\widehat{\mathbb{E}}_a \widehat{T}_{\hat{c}} = \widehat{M}$, and $\widehat{M}/M \to N + 1$ by Proposition 10(iii). □

The program `merge_and_comb_checks.py` in the repository named below checks, in rational arithmetic, the formula for $\widehat{F}$ and inequality (4) for every pair of states and every lumping of two states in 100 random networks with three to six states (850 lumpings), and the closed forms of Proposition 10 for $N \le 3$ and $t \in \{2, 3, 10, 100, 1000\}$. The program `lumping_checks.py` checks Proposition 4, the inequalities $\tau_1^{(2)}(\widehat{X}) \le \tau_0(\widehat{X}) \le \tau_0(X) \le (m-1)\,\tau_1^{(2)}(X)$ and the upper bound of Theorem 1 on 400 random networks with three to six states and random partitions.

Acknowledgements and funding

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [7] and working together inside the Brain framework [9], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [8], the Math Brain repository [7] and the Brain framework repository [9].

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [9] and Math Brain [7]. In October 2026 an AI agent working from the project's list of open problems found examples in which lumping increases $\tau_1^{(2)}$. A Claude Fable agent (Anthropic) obtained Lemma 3 and Proposition 4, ran the programs and prepared the manuscript. A GPT model (OpenAI) found the bound $K(n) \leq n - 2$, and Codex agents (OpenAI) found Theorem 2 and the comb of Section 5 and recomputed the examples with their own programs. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Programs and outputs:
`https://github.com/yspennstate/lumping-mixing-time`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [7].

Appendix A. Notation

Symbols are listed in the order in which they first appear. Letters used only inside one proof are not listed.

Symbol	Meaning	First appears
I, n	the state space of X and its number of elements	Section 1
w_{xy}, w_x, w	the weight of $\{x, y\}$, the weight of a state, the total weight	Section 1
X, p_{xy}, π	the random walk, its transition probabilities, its stationary law	Section 1
T_y, T_A	the hitting time of a state y and of a set A	Section 1
$\mathbb{E}_\mu$, $\mathbb{P}_\mu$	expectation and probability for the chain started from μ	Section 1
f, $\widehat{I}$	the lumping map and its image set	Section 1
$\widehat{X}$, $\widehat{w}$, $\widehat{\pi}$	the induced chain, its weights, its stationary law	Section 1
$\widehat{\mathbb{E}}$, $\widehat{T}$	expectation and hitting times for $\widehat{X}$	Section 1
$\tau_1^{(2)}$	the stopping-rule mixing time, given by (1)	Section 1
τ^*, τ_0	the maximal mean commute time and the average hitting time	Section 1
λ_k, τ_2	the k-th largest eigenvalue of (p_{xy}) and $1/(1-\lambda_2)$	Section 1
τ_c	the Cheeger time constant	Section 1
$K(n)$	the largest factor by which lumping can increase $\tau_1^{(2)}$ on at most n states	Section 1
$U(n)$	the upper bound $\max_q \min(2^{n-q}, q-1)$	Section 1
m, q	the numbers of states of X and of $\widehat{X}$	Section 1
$H(x,j)$, F_{ij}	$\mathbb{E}_x T_j$ and $\mathbb{E}_i T_j - \mathbb{E}_\pi T_j$	Section 2
ω, $\mathbf{1}$, δ_x	the vector (w_x), the constant 1, the indicator of x	Section 2
L, $\mathcal{E}$	the Laplacian (loops removed) and the Dirichlet form	Section 2
$R(y \leftrightarrow Z)$	the effective resistance between y and Z	Section 2
o, Γ	the grounded state and the grounded inverse of L	Section 2
ζ, $\bar{\zeta}$	$\Gamma\omega$, so $\zeta_i = H(i,o)$, and its π-mean	Section 2
κ_i, φ_i	$\max_j F_{ij}$ and the potential $\kappa_i - F_i$.	Lemma 5
V_{xy}	the states reached from x without visiting y	Lemma 6
S_e, w_e	the weight beyond an edge e of a tree, and the weight of e	Lemma 7
u, v, J, $\widehat{F}$	the lumped pair, the other states, the matrix F of $\widehat{X}$	Section 3
h, R, $\bar{h}$	$\mathbb{P}_x(T_u < T_v)$, $R(u \leftrightarrow v)$, the π-mean of h	Section 3
β_i	$\varphi_i(u) - \varphi_i(v) = wR(h_i - \bar{h})$	Section 3
η, $\bar{\eta}$	the pointwise minimum of φ_i/β_i and φ_j/β_j, and its π-mean	Section 3
π_*	$\min_x \pi_x$	Remark 9
N, t, r, α, ℓ	the parameters of the comb	Section 5
a, b, c, z_k, y_k	the states of the comb: two ends, the heavy leaf, spine, teeth	Section 5
$\hat{z}_k$, $\hat{c}$	the lumped states $f(z_k) = f(y_k)$ and $f(z_N) = f(c)$	Section 5
σ_k, B, S_k, $\widehat{S}_k$	the constants defined before Proposition 10	Section 5
M, $\widehat{M}$	$\mathbb{E}_a T_c$ in X and $\widehat{\mathbb{E}}_a \widehat{T}_{\hat{c}}$ in $\widehat{X}$	Section 5
$G_Z(x,y)$	the expected number of visits to y before T_Z from x	Section 6

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